Predictive Skill-Seam Action Models for Robot Composition

One-page collaboration memo for Paper 119

Core idea: treat skill handoff as a local world/action model: predict whether composition will fail, diagnose why, choose repair/probe/abstain/transition, and update future planning.

Problem. Robot skills that work individually can fail when composed. A graph edge may look legal while the terminal state of the first skill falls outside the next skill's attraction basin, crosses a high-energy barrier, or enters a contact mode where the next controller no longer descends. The failure is not only low-level control; it is a representation and prediction problem at the action seam.

Method. The proposed composer scores candidate handoff h by basin overlap, barrier height, descent continuity, repair cost, and calibrated seam risk:

$$S(i, j, h) = \alpha O_{\text{basin}} - \beta B_{\text{barrier}} + \gamma D_{\text{descent}} - \lambda C_{\text{repair}} - \kappa R_{\text{seam}}.$$

It then chooses among **accept**, **repair**, **probe**, **abstain**, or **alternate transition**. The contribution is a seam-level predictive interface in the world/action-modeling family, not a new low-level controller or universal manipulation policy.

Current evidence.

Hard-slice metric	Proposed	Strongest non-oracle predecessor
Success	0.80171	0.71711
Utility	0.88827	0.65310
Paired utility wins	10/10	–

The frozen local suite covers 12 methods, 6 task families, 8 seam regimes, 5 deployment splits, and 10 paired seeds. The proposed method reduces seam failure, barrier violation, damage, risk calibration error, and realized seam breach while improving basin alignment and descent continuity. A diagnostic audit checks exported failure labels, seam decisions, and planner-edge updates over 230,400 local rows.

Claim boundary. The current claim is bounded: energy-seam certification improves skill composition when basin, barrier, descent, repair, and risk quantities are estimable. The paper should not claim deployment-ready robotics until it has real-robot or accepted high-fidelity validation with shared skill libraries, paired resets, raw logs, videos, and independent baselines.

Why this is a natural fit for behavior composition. CoStream asks how simple behaviors can compose into complex manipulation. Paper 119 asks when a composed behavior handoff should be trusted, repaired, probed, or avoided. A clean collaboration question is whether a seam model improves robustness for a CoStream-style stack under perturbations, precision assembly, object transfer, and contact-mode transitions.

Independent validation packet. Use the same primitive skills across baselines; log terminal samples, basin estimates, barrier scores, descent diagnostics, predicted risk, accepted/rejected seams, outcomes, and videos. The repo now has a no-go operator packet that lists the remaining backend, config, fidelity, alias-unsealing, and run-id gates before any external collection starts; it is validation scaffolding, not evidence.

References for fit: Haonan Chen homepage; CoStream; Yilun Du homepage; SIMPACT.